# New Reactions

1. **Acidity of Amines and Amide Formation (Section 21-4)**

$$RNH_2 + H_2O \overset{K}{\rightleftharpoons} R\overset{-}{N}H + H_3O^+ \quad K_a \approx 10^{-35}$$

$$\underset{}{R_2NH} + CH_3CH_2CH_2CH_2Li \rightleftharpoons \underset{\textbf{Lithium dialkylamide}}{R_2N^-Li^+} + CH_3CH_2CH_2CH_3$$

$$2\ NH_3 + 2\ Na \xrightarrow{\text{Catalytic } Fe^{3+}} 2\ NaNH_2 + H_2$$

2. **Basicity of Amines (Section 21-4)**

$$RNH_2 + H_2O \rightleftharpoons R\overset{+}{N}H_3 + HO^-$$

$$R\overset{+}{N}H_3 + H_2O \overset{K}{\rightleftharpoons} RNH_2 + H_3O^+ \quad K_a \approx 10^{-10}$$

Salt formation

$$RNH_2 + HCl \longrightarrow \underset{\textbf{Alkylammonium chloride}}{R\overset{+}{N}H_3\ Cl^-}$$

General for primary, secondary, and tertiary amines

**Preparation of Amines**

3. **Amines by Alkylation (Section 21-5)**

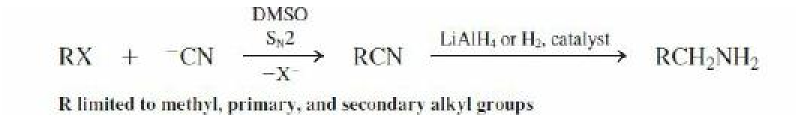
$$R\ddot{N}H_2 + R'X \longrightarrow R\overset{\overset{R'}{|}}{\overset{+}{N}}H_2\ X^-$$

General for primary, secondary, and tertiary amines

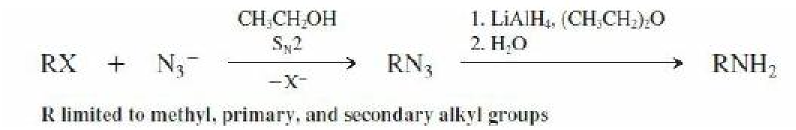
Drawback: multiple alkylation

$$R\overset{\overset{R'}{|}}{\overset{+}{N}}H_2\ X^- + R'X \longrightarrow \longrightarrow \longrightarrow R\overset{+}{N}R'_3\ X^-$$

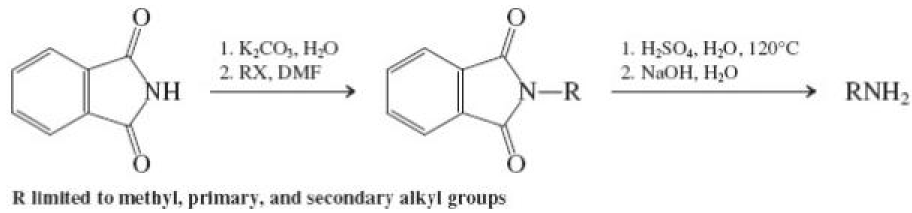
4. **Primary Amines from Nitriles (Section 21-5)**



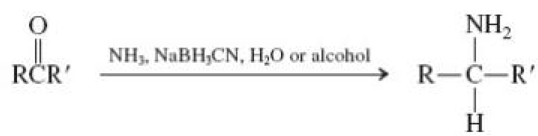

## 5. Primary Amines from Azides (Section 21-5)



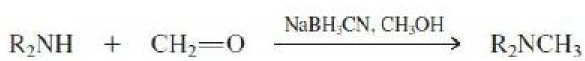

## 6. Primary Amines by Gabriel Synthesis (Section 21-5)



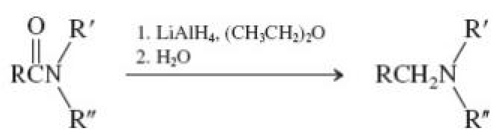

## 7. Amines by Reductive Amination (Section 21-6)



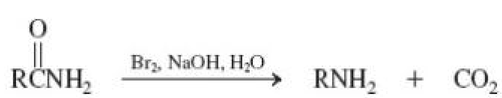

Reductive methylation with formaldehyde




## 8. Amines from Carboxylic Amides (Section 21-7)




## 9. Hofmann Rearrangement (Section 21-7)



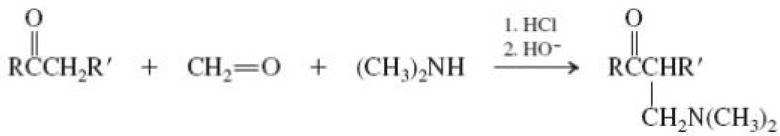
## Reactions of Amines

### 10. Hofmann Elimination (Section 21-8)

$$RCH_2CH_2NH_2 \xrightarrow{\text{Excess } CH_3I,\ K_2CO_3} RCH_2CH_2\overset{+}{N}(CH_3)_3\ I^- \xrightarrow[-AgI]{Ag_2O,\ H_2O}$$

$$RCH_2CH_2\overset{+}{N}(CH_3)_3\ {}^-OH \xrightarrow{\Delta} RCH{=}CH_2 + N(CH_3)_3 + H_2O$$

### 11. Mannich Reaction (Section 21-9)

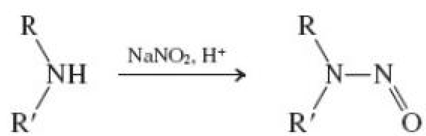


### 12. Nitrosation of Amines (Section 21-10)

Tertiary amines

$$R_3N \xrightarrow{NaNO_2,\ H\text{-}X^-} \underset{\textbf{Tertiary } \textbf{\textit{N}}\textbf{-nitrosammonium salt}}{R_3\overset{+}{N}{-}NO\ X^-}$$

Secondary amines

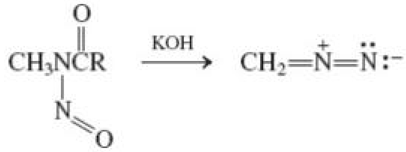


***N*-Nitrosamine**

Primary amines

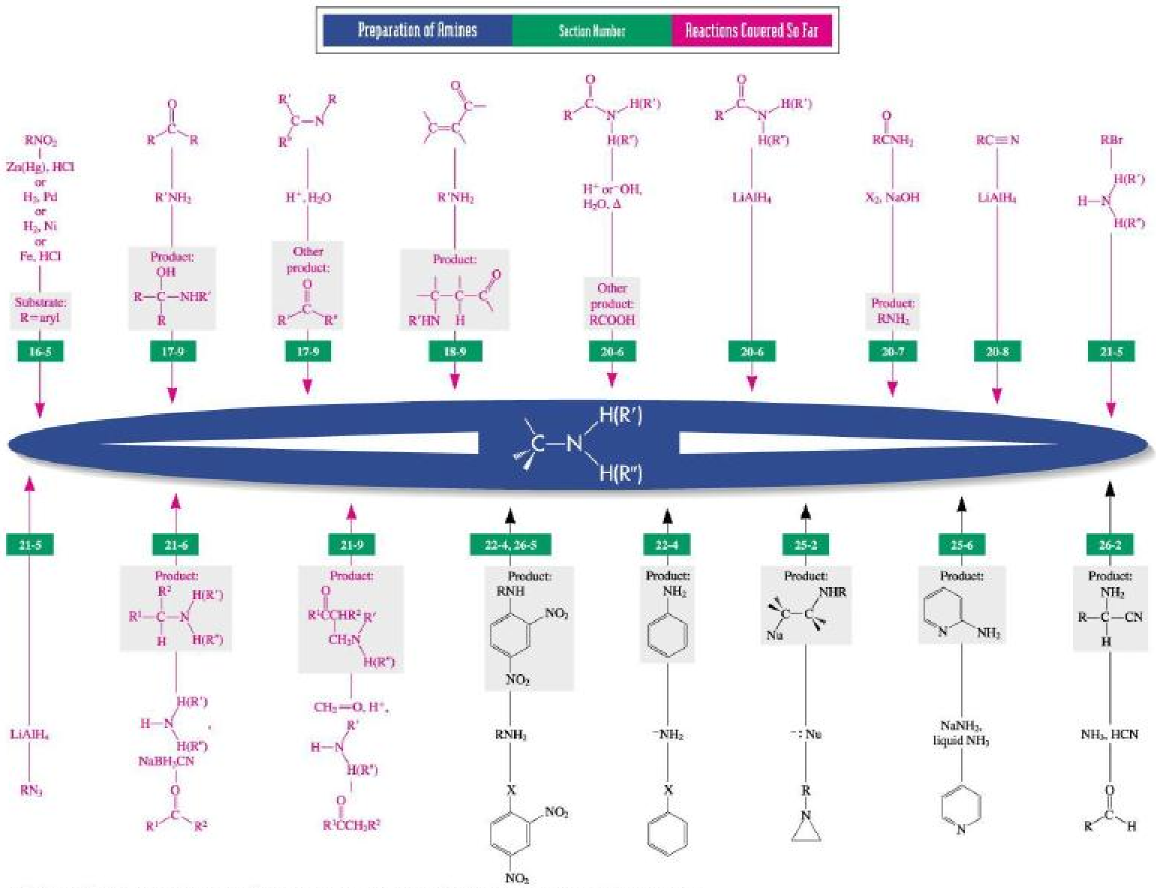
$$RNH_2 \xrightarrow{NaNO_2,\ H^+} RN{=}NOH \xrightarrow[-H_2O]{H^+} RN_2^+ \xrightarrow[-N_2]{} R^+ \longrightarrow \text{mixture of products}$$

### 13. Diazomethane (Section 21-10)



Esterification with diazomethane

$$\mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}OH} \;+\; \mathrm{CH_2N_2} \;\longrightarrow\; \mathrm{R\overset{\overset{\displaystyle O}{\|}}{C}OCH_3} \;+\; \mathrm{N_2}$$

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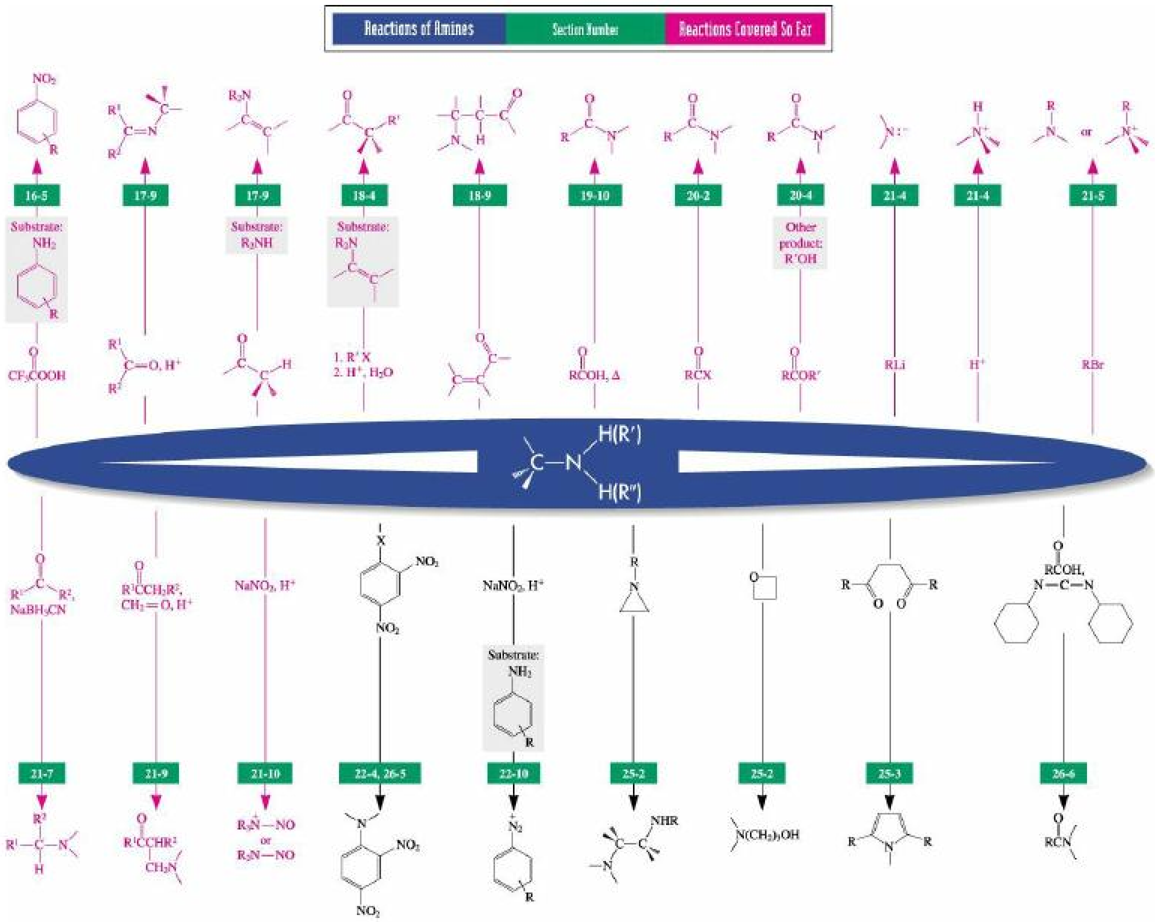
Reactions of Amines
Section Number
Reactions Covered So Far
16-5
17-9
17-9
18-4
18-9
19-10
20-2
20-4
21-4
21-4
21-5
Substrate:
Substrate:
Substrate:
Other product:
R′OH
CF3COOH
H+
RLi
RBr
NaNO2, H+
NaNO2, H+
Substrate:
21-7
21-9
21-10
22-4, 26-5
22-10
25-2
25-2
25-3
26-6

# Important Concepts

1. **Amines** can be viewed as derivatives of **ammonia,** just as ethers and alcohols can be regarded as derivatives of water.

2. *Chemical Abstracts* names amines as **alkanamines** (and **anilines),** alkyl substituents on the nitrogen being designated as *N*-alkyl. Another system is based on the label aminoalkane. Common names are based on the label alkylamine.

3. The **nitrogen** in amines is **sp3**$sp^3$ **hybridized,** the nonbonding electron pair functioning as the equivalent of a substituent. This tetrahedral arrangement **inverts rapidly** through a planar transition state.

4. The **lone electron pair** in amines is less tightly held than in alcohols and ethers, because nitrogen is **less electronegative** than oxygen. The consequences are a diminished capability for hydrogen bonding, higher basicity and nucleophilicity, and lower acidity.

5. **Infrared spectroscopy** helps to differentiate between primary and secondary amines. **Nuclear magnetic resonance** spectroscopy indicates the presence of nitrogen-bound hydrogens; both hydrogen and carbon atoms are **deshielded** in the vicinity of the nitrogen. **Mass spectra** are characterized by **iminium ion** fragments.

6. Indirect methods, such as displacements with azide or cyanide, or reductive amination, are superior to direct alkylation of ammonia for the **synthesis** of amines.

7. The NR3$\mathbf{NR_3}$ group in a quaternary amine, $R'-\overset{+}{N}R_3$, is a **good leaving group** in E2 reactions; this enables the Hofmann elimination to take place.

8. The **nucleophilic reactivity** of amines manifests itself in reactions with electrophilic carbon, as in haloalkanes, aldehydes, ketones, and carboxylic acids and their derivatives.